VIGNAN'S INSTITUTE OF INFORMATION TECHNOLOGY (AUTONOMOUS)
(Approved by AICTE - New Delhi & Affiliated to JNTU-GV, Vizianagaram)
Beside VSEZ, Duvvada, Vadlapudi Post, Gajuwaka, Visakhapatnam - 530 049.

# A REPORT

On

# Community Service Project

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**Name and Address of the Habitation for Community Service Project (CSP):**

Dungalavanipalem, NAD Junction, Visakhapatnam, Andhra Pradesh

# Community Service Project Report

on

# SachiSeva – Welfare Eligibility Platform

*Submitted in partial fulfillment of the requirements for the award of the Community Service Project of*

## Bachelor of Technology

In

**Department of Computer Science Engineering-Artificial Intelligence**

By


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Under the Faculty Guidance of

**Mrs. K. Swathi**

**Assistant Professor & Head of the Department**




**Department of Computer Science Engineering-Artificial Intelligence**

**Vignan's Institute of Information Technology(A)**


**June, 2026**

**Duvvada, Visakhapatnam**

# Vignan's Institute of Information Technology(A)

**Department of Computer Science Engineering-Artificial Intelligence**

# STUDENT DECLARATION

We, **B. Maithreya** (Reg. No: **24L31A4318**), **D. Abhilash** (Reg. No: **24L31A4340**), **J. Mounika** (Reg. No: **24L31A4357**), and **K. Yogitha** (Reg. No: **24L31A4359**), students of II B.Tech in the Department of Computer Science & Engineering – AI, Vignan's Institute of Information Technology (A), Duvvada, Visakhapatnam, hereby declare that the presented report of the Community Service Project titled "**SachiSeva – Welfare Eligibility Platform**" is uniquely prepared by us after successful completion of a summer Community Service Project from 11 May 2026 to 20 June 2026 in Dungalavanipalem, NAD Junction, under the faculty guidance of Mrs. K. Swathi, Assistant Professor & Head of the Department, Department of Computer Science & Engineering – AI, Vignan's Institute of Information Technology (A), Duvvada, Visakhapatnam, during the academic year 2025-2026.

We also confirm that the report is only prepared for our academic requirement, not for any other purpose. It might not be used in the interest of the opposite party of the corporation.

**Name of Students:** B. Maithreya, D. Abhilash, J. Mounika, K. Yogitha

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# CERTIFICATE

This is to certify that **B. Maithreya** (24L31A4318), **D. Abhilash** (24L31A4340), **J. Mounika** (24L31A4357), and **K. Yogitha** (24L31A4359) have completed the Community Service Project at Dungalavanipalem, NAD Junction on "**SachiSeva – Welfare Eligibility Platform**" from 11 May 2026 to 20 June 2026 under the faculty guidance of Mrs. K. Swathi, Assistant Professor & Head of the Department. They have also submitted the Community Service Project report to the department during the academic year 2025-2026, in partial fulfilment of the requirements for the award of the Community Service Project of Bachelor of Technology in Computer Science & Engineering – AI, Vignan's Institute of Information Technology (A), Duvvada, Visakhapatnam.

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# ABSTRACT

SachiSeva – Welfare Eligibility Platform is a bilingual (Telugu and English) web application developed as a Community Service Project for the citizens of Andhra Pradesh. It brings the major State and Central welfare schemes and the common Grama/Ward Sachivalayam services into one place, and lets a family find out, in one click, every scheme it qualifies for, the benefit it would receive and the documents it needs.

The app addresses a significant gap in welfare outreach: many eligible citizens are unaware of, or unable to access, benefits because of language barriers, limited digital literacy and a lack of guidance on documents. SachiSeva resolves this with a Telugu/English interface, a family profile that is answered only once, an eligibility engine based on the government's six-step validation, one combined document checklist, an "Apply for all" option, voice read-out and voice input, and an assistant (SachiBot) that answers in Telugu or English.

The application is built with React, TypeScript and Tailwind CSS, uses Supabase (a PostgreSQL database with row-level security, sign-in, document storage and live updates) as its backend, and is hosted on Vercel. It covers 21 welfare schemes and 16 Sachivalayam services in 14 categories. Citizens apply online, upload each document once, track every application and message Sachivalayam staff; staff review applications in a separate panel that needs a staff key, and when an application is approved or rejected the applicant is automatically asked to visit their nearby Sachivalayam for the next steps.

The system was validated with 65 automated database checks covering every role and workflow, and with automated browser tests of every screen in Telugu and English on phone and desktop layouts.

# ACKNOWLEDGEMENTS

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**B. Maithreya, D. Abhilash, J. Mounika, K. Yogitha**

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# CHAPTER 1: EXECUTIVE SUMMARY

Many citizens of Andhra Pradesh do not get the welfare benefits they are entitled to, not because they are ineligible but because they do not know which schemes apply to them, what each scheme gives, or which documents they must carry. Scheme information is spread across government notices and portals, and is usually obtained by asking Sachivalayam staff about one scheme at a time.

To address this, our team developed **SachiSeva – Welfare Eligibility Platform** as part of the Community Service Project at Dungalavanipalem, NAD Junction. It is a bilingual (Telugu/English) web application where a family answers a few questions once and instantly sees every scheme it qualifies for, with the benefit amount and the documents required. The family can then apply for all of those schemes together, upload each document only once, and follow every application until Sachivalayam staff decide on it.

**Activities Performed:**

- Surveyed residents and Sachivalayam visitors at Dungalavanipalem, NAD Junction on scheme awareness, preferred language and repeat visits caused by missing documents.
- Built a catalogue of 21 welfare schemes and 16 Sachivalayam services in 14 categories, verified against current government sources.
- Designed an eligibility engine based on the government's six-step check that gives eligible, likely eligible and not eligible results.
- Developed the citizen application with a combined document checklist, "Apply for all", application tracking, voice read-out and the SachiBot assistant.
- Developed a staff panel, protected by a staff key, to review applications, request documents and record decisions.
- Conducted awareness sessions and demonstrated the application to citizens on their phones.

**The Learning Objectives and Outcomes Achieved:**

**Technical Understanding:** Gained practical knowledge of full-stack web development with React and TypeScript, a PostgreSQL database with row-level security on Supabase, server functions on Vercel, and the Web Speech API for Telugu voice read-out.

**System Design and Planning:** Improved skills in modelling scheme data, designing a rule-based eligibility engine, planning secure roles for citizens and staff, and testing every workflow with automated database and browser tests.

**Social Responsibility:** Understood the welfare-scheme ecosystem of the AP Sachivalayam system and how a simple, bilingual tool can help citizens with limited reading ability reach the benefits meant for them.

# CHAPTER 2: OVERVIEW OF THE COMMUNITY

## Introduction

The Community Service Project was carried out at Dungalavanipalem, NAD Junction, Visakhapatnam, Andhra Pradesh, a settlement served by a local Grama/Ward Sachivalayam. Before finalising the scope of SachiSeva, an informal survey was carried out among residents and visitors at the local Sachivalayam to confirm that the welfare-scheme awareness gap was a genuine, everyday problem. Respondents were asked how many schemes they could name that they were eligible for, which language they preferred, whether a missing document had caused a repeat visit, and whether they would use a phone-based tool in Telugu.

**Table 1** – Survey Respondent Summary

| Respondent | Age Group | Schemes Correctly Named | Preferred Language |
|---|---|---|---|
| Ramya | 18–29 | 3 | English |
| Vinay | 30–45 | 2 | Telugu |
| Raju | 46–60 | 4 | Telugu |
| Prem | 60+ | 1 | Telugu |

The survey indicated that respondents could typically name far fewer schemes than they were actually eligible for, that a majority preferred Telugu for reading scheme information, and that a notable share reported a repeat Sachivalayam visit caused by a missing document.

## Target Audience

**SachiSeva – Welfare Eligibility Platform benefits:**

1. **Citizens and families** – one place to see every scheme they qualify for, with benefits and documents.
2. **Elderly citizens and those with limited reading ability** – Telugu screens and a speaker button that reads every page aloud.
3. **Sachivalayam staff** – a panel to review applications, request documents and record decisions.

4. **The community** – better awareness of welfare schemes and fewer wasted trips to the office.

### Community Impact

Dungalavanipalem, NAD Junction faces challenges common to communities served by the Sachivalayam system: dependence on staff and notice boards for scheme information, limited awareness of eligibility rules that change whenever a scheme is revised, and difficulty with formal Telugu/English terminology. Mobile phone ownership among respondents was high, though comfort with reading detailed government notices was uneven, particularly among older citizens.

### Ethical Challenges

The project handles personal and family information, so privacy was treated as a requirement. Only the last four digits of Aadhaar are collected, each citizen can see only their own profile, applications and documents, and eligibility results are presented as guidance, with the final decision left to Sachivalayam staff.

### Digital Divide

- Citizens familiar with smartphones can check their schemes and apply on their own.
- Citizens with limited reading ability can listen to each page in Telugu, and staff can guide them at the Sachivalayam.

### Role of the Project in the Community

**Table 2** – Role of the Project in the Community

| Role | Description |
|---|---|
| **Awareness** | Shows each family every scheme it qualifies for, with the benefit amount. |
| **Accessibility** | Telugu/English screens and voice read-out for citizens with limited reading ability. |
| **Convenience** | One combined document list and "Apply for all" instead of separate applications. |
| **Transparency** | Live status of each application and messages from Sachivalayam staff. |

# CHAPTER 3: COMMUNITY SERVICE PART

As part of our community service activities, we studied how citizens at Dungalavanipalem, NAD Junction learn about welfare schemes and found that most of them depend on asking Sachivalayam staff about one scheme at a time. To address this, we developed SachiSeva – Welfare Eligibility Platform, a bilingual web application that shows a family every scheme it qualifies for and lets it apply for all of them together.

### Activities Undertaken Community Interaction:

We interacted with residents and visitors at the local Sachivalayam to understand how they find out about schemes, which language they prefer, and what problems they face when applying. Many said they had made extra trips because a document was missing.

### Idea Collection:

Based on the feedback, we identified the need for a single place where a family can check all its schemes in Telugu, see one combined list of documents, and apply without repeated visits.

### What We Observed:

Citizens could name only a few of the schemes they were eligible for, preferred Telugu, and often did not know which documents to bring until they reached the office.

### What We Decided to Do:

We decided to build a bilingual web application with a one-time family questionnaire, an eligibility engine based on the government's six-step check, a combined document checklist, online applications with tracking, and a staff panel.

### Content Preparation:

We prepared the catalogue of 21 schemes and 16 services in Telugu and English, verified benefit amounts and documents against government sources, built the citizen and staff screens, and tested every workflow.

### Awareness Demonstration:

We demonstrated SachiSeva to citizens at the Sachivalayam, showing how to answer the family questions once, see eligible schemes and benefits, listen to the required documents in Telugu, and apply for all eligible schemes together.

**Table 3** – Community Service Activities Undertaken

| Activity | Summary |
|---|---|
| **Community Interaction** | Spoke with residents and Sachivalayam visitors about how they learn about schemes and the problems they face. |
| **Idea Collection** | Identified the need for one Telugu-first place to check all schemes, documents and applications. |
| **What We Observed** | Citizens knew few of their schemes and often made repeat visits because of missing documents. |
| **What We Decided to Do** | Build a bilingual eligibility, application and tracking app with a staff panel. |
| **Content Preparation** | Prepared and verified the catalogue of 21 schemes and 16 services, and built and tested the app. |
| **Awareness Demonstration** | Demonstrated SachiSeva to citizens at the Sachivalayam on their phones. |

# ACTIVITY LOG FOR WEEK-1

| Day | Date | Brief description of the daily activity | Learning Outcome | Person In-Charge Signature |
|---|---|---|---|---|
| Day-1 | May 11, 2026 | Team formation and project ideation; identified the welfare-scheme awareness gap in the AP Sachivalayam system as the focus area. | Understood project objectives and teamwork. | |
| Day-2 | May 12, 2026 | Studied the AP Sachivalayam welfare-scheme ecosystem and the challenges citizens face in identifying schemes they qualify for. | Awareness of real-world governance problems. | |
| Day-3 | May 13, 2026 | Reviewed bilingual e-governance tools and web-based welfare-eligibility services. | Research and analytical skills. | |
| Day-4 | May 14, 2026 | Identified requirements: Telugu/English interface, eligibility engine, online applications and a staff review panel. | Project planning skills. | |
| Day-5 | May 15, 2026 | Studied React, Supabase (database, sign-in, file storage) and Web Speech API concepts. | Understood technology fundamentals. | |
| Day-6 | May 16, 2026 | Finalized project scope, objectives and the development roadmap for SachiSeva. | Requirement analysis and documentation skills. | |

# WEEKLY REPORT

## WEEK – 1 (From Date: May 11 to Date: May 16)

**Objective of the Activity Done:**

To initiate the SachiSeva project, understand the problem domain of welfare-scheme awareness, and define the project objectives and requirements.

**Detailed Report:**

During the first week of the project, our team formed, divided roles for research, design, development and field work, and agreed on a weekly working schedule. We identified the welfare-scheme awareness gap in the AP Sachivalayam system as the focus of our Community Service Project.

We studied how citizens currently find out about AP welfare schemes through the Grama/Ward Sachivalayam and listed the main difficulties they face. Existing government portals and e-governance applications were reviewed to understand what information they provide and where citizens get stuck.

We then wrote down the core requirements: a Telugu/English interface, a one-time family questionnaire, an eligibility check across all schemes, online applications and a staff review panel. The chosen technologies were studied: React for the web application, Supabase for the database, sign-in and file storage, and the Web Speech API for voice read-out.

Because many citizens are more comfortable in Telugu than in English, bilingual text and voice read-out were made core requirements. By the end of the week the team had a clear problem statement, an agreed list of features and a technology plan.

# ACTIVITY LOG FOR WEEK-2

| Day | Date | Brief description of the daily activity | Learning Outcome | Person In-Charge Signature |
|---|---|---|---|---|
| Day-7 | May 18, 2026 | Designed the system architecture: React web application, Supabase backend and Vercel server functions. | System architecture design skills. | |
| Day-8 | May 19, 2026 | Modelled scheme data: eligibility rules, benefits, documents, steps, fees and time limits in Telugu and English. | Data modelling skills. | |
| Day-9 | May 20, 2026 | Identified and catalogued welfare schemes and citizen services across multiple categories. | Data source analysis. | |
| Day-10 | May 21, 2026 | Designed the eligibility-matching engine (profile compared against scheme rules). | Rule-engine design skills. | |
| Day-11 | May 22, 2026 | Designed the bilingual (Telugu/English) screens and navigation flow. | UI/UX design skills. | |
| Day-12 | May 23, 2026 | Planned data security: row-level security for citizens and a separate staff role. | Database security skills. | |

# WEEKLY REPORT

## WEEK – 2 (From Date: May 18 to Date: May 23)

**Objective of the Activity Done:**

To design the SachiSeva platform: system architecture, data model, eligibility engine and bilingual screens.

**Detailed Report:**

In Week 2, our focus shifted to designing the platform. On Day 7, the architecture was finalized: a React single-page application hosted on Vercel, Supabase for the database, sign-in, document storage and live updates, and small Vercel server functions for the assistant and centre search.

On Days 8 and 9, the scheme record was designed with the name in Telugu and English, category, benefit, eligibility rules, required documents, application steps, fees and time limits, and 21 schemes and 16 services were catalogued across 14 categories.

On Day 10, the eligibility engine was designed so that one set of family answers is matched against every scheme at once. Because some rules cannot be decided from a short questionnaire, it gives three results: eligible, likely eligible and not eligible, so that citizens are not wrongly turned away.

Days 11 and 12 focused on the main screens and on security: citizens see only their own applications, staff accounts need a staff key, and only the last four digits of Aadhaar are collected. By the end of the week, the complete design of SachiSeva was ready.

# ACTIVITY LOG FOR WEEK-3

| Day | Date | Brief description of the daily activity | Learning Outcome | Person In-Charge Signature |
|---|---|---|---|---|
| Day-13 | May 25, 2026 | Prepared a survey questionnaire for Dungalavanipalem, NAD Junction on scheme awareness, language preference and document issues. | Survey design and research skills. | |
| Day-14 | May 26, 2026 | Visited Dungalavanipalem, NAD Junction and interacted with citizens and Sachivalayam visitors. | Community engagement and communication. | |
| Day-15 | May 27, 2026 | Collected survey responses from citizens on welfare-scheme awareness and access. | Data collection and field research skills. | |
| Day-16 | May 28, 2026 | Conducted awareness sessions explaining the SachiSeva eligibility-checking tool. | Public awareness and presentation skills. | |
| Day-17 | May 29, 2026 | Demonstrated the SachiSeva application on mobile devices to citizens. | Technical demonstration and user education. | |
| Day-18 | May 30, 2026 | Analyzed the survey responses and identified key awareness gaps in the community. | Data analysis and problem identification. | |

# WEEKLY REPORT

## WEEK – 3 (From Date: May 25 to Date: May 30)

**Objective of the Activity Done:**

To conduct the community survey at Dungalavanipalem, NAD Junction and create awareness about the SachiSeva platform among citizens.

**Detailed Report:**

The third week was dedicated to community engagement. On Day 13, we prepared a short survey on scheme awareness, preferred language, source of scheme information and repeat visits caused by missing documents.

On Days 14 and 15, we visited Dungalavanipalem, NAD Junction, spoke with citizens and Sachivalayam visitors, and recorded their responses. Common questions were about pensions, rice cards, certificates and education support.

On Days 16 and 17, awareness sessions showed how to answer the family questions once and see all eligible schemes with their document lists, and the application was demonstrated on mobile phones, including the Telugu interface and voice read-out. Some citizens were hesitant to share personal details, so we explained that only the last four digits of Aadhaar are asked for.

On Day 18, the responses were analyzed. The survey confirmed that citizens usually learn about schemes one at a time and often make extra trips because of missing documents, and they responded well to seeing all their schemes and one document checklist on a single screen in Telugu.

# ACTIVITY LOG FOR WEEK-4

| Day | Date | Brief description of the daily activity | Learning Outcome | Person In-Charge Signature |
|---|---|---|---|---|
| Day-19 | June 1, 2026 | Implemented and tested the eligibility engine against representative citizen profiles. | Testing and validation skills. | |
| Day-20 | June 2, 2026 | Verified scheme names, benefit amounts and document checklists against current government sources. | Data verification skills. | |
| Day-21 | June 3, 2026 | Tested online applications, document upload, the staff panel, voice read-out and SachiBot. | Quality assurance skills. | |
| Day-22 | June 4, 2026 | Conducted a follow-up awareness session and gathered citizen feedback. | Leadership and awareness creation. | |
| Day-23 | June 5, 2026 | Deployed the application and demonstrated the final SachiSeva tool to the community. | Technical presentation skills. | |
| Day-24 | June 6, 2026 | Compiled the survey findings and prepared the final community analysis report. | Data synthesis and reporting skills. | |

# WEEKLY REPORT

## WEEK – 4 (From Date: June 1 to Date: June 6)

**Objective of the Activity Done:**

To complete testing of the SachiSeva application, gather citizen feedback, and prepare the final community analysis report.

**Detailed Report:**

Week 4 focused on testing and finalizing SachiSeva. On Day 19, the eligibility engine was tested with sample family profiles covering farmers, students, elderly citizens, widows and auto drivers.

On Day 20, scheme names, benefit amounts and document lists were checked against current official AP government sources and outdated entries were corrected. Each scheme was marked as open, announced or not currently accepting applications, so that citizens are not misled.

On Day 21, the full flow was tested: sign-up, the family questionnaire, applying online, uploading documents, tracking applications and staff review, together with the nearest-centre map, the Telugu voice read-out and SachiBot on mobile phones.

On Days 22 and 23, a follow-up session was held with citizens, their feedback was collected, and the deployed application was demonstrated to the community. Finally, on Day 24, the survey findings were compiled into the final report.

# CHAPTER 5: OUTCOMES DESCRIPTION

**Details of the Survey**

As part of the Community Service Project, a survey was conducted at Dungalavanipalem, NAD Junction to understand citizens' awareness of AP government welfare schemes, the barriers they face in accessing entitlements, and their preferred language for scheme information (see Table 1). The survey revealed that respondents could typically name far fewer schemes than they were actually eligible for, learnt about schemes mainly by asking staff about one scheme at a time, and often made repeat visits because a document was missing. A majority preferred Telugu. These findings highlight the need for a bilingual, phone-friendly, self-service eligibility tool like SachiSeva.

**Problems Identified in the Community**

- Citizens learn about a scheme only by asking Sachivalayam staff about one scheme at a time.
- Eligibility rules are spread across many notices and change whenever a scheme is revised.
- Scheme information is often in formal Telugu or English, hard for citizens with limited literacy.
- Citizens make repeat visits to the Sachivalayam because a required document is missing.
- No single, simple place exists to check, in one's own language, exactly which schemes apply.

**Short-Term Action Plan**

SachiSeva was demonstrated to citizens visiting the Dungalavanipalem, NAD Junction Sachivalayam, who were helped to add it to their phones. Digital-literacy sessions showed citizens how to use the family-profile eligibility check on their own, awareness material with the SachiSeva link was proposed for the Sachivalayam, and citizen feedback was collected to refine the wording of questions and document checklists.

### Long-Term Action Plan

In the long term, the scheme catalogue should be extended as new schemes are announced and re-verified against official sources, push notifications should tell citizens about application updates and renewals, a larger user study should measure the time saved per enquiry, and, with permission, the app should connect to official data sources so that eligibility and application status are confirmed automatically.

### Community Awareness Programmes Conducted

Awareness programmes were conducted at the Sachivalayam with interactive sessions, live demonstrations of SachiSeva and discussions about the documents required for common schemes. Citizens were shown how to answer the family questions once, view their personalised list of eligible schemes with benefit amounts, use the voice read-out to hear the required documents, and apply for all their eligible schemes together.

### Key Outcomes

- Citizens gained a better understanding of the welfare schemes they were personally eligible for.
- Awareness of document requirements improved, which can reduce repeat Sachivalayam visits.
- Citizens with limited reading ability showed interest in the Telugu voice read-out.
- SachiSeva was well received as a practical and accessible solution for the community.

**Table 4** – Summary of Key Outcomes by Area

| Outcome Area | Key Outcomes |
|---|---|
| **Scheme Awareness** | Each family sees every scheme it qualifies for, with the personalised benefit amount. |
| **Document Preparation** | One combined checklist for all eligible schemes; each document is uploaded once. |
| **Applications and Tracking** | "Apply for all" creates an application for every scheme, with live status and staff messages. |
| **Accessibility** | Telugu/English screens and a speaker button on every page. |
| **Data Security** | Each citizen sees only their own data; only the last four Aadhaar digits are stored. |

# REPORT OF THE MINI-PROJECT WORK SACHISEVA – WELFARE ELIGIBILITY PLATFORM

## 1. Introduction

The Government of Andhra Pradesh delivers a large share of its welfare benefits through the Grama/Ward Sachivalayam (village/ward secretariat) system. The number of active welfare schemes has grown steadily and spans pensions, education, agriculture, housing, health, support for weavers, fishermen, women, youth and SC/ST/BC/minority communities, and citizen document services such as caste, income and residence certificates. Under the present government, several of these schemes were re-branded or re-launched, including six flagship promises known as the "Super Six".

Although information about these schemes exists in circulars, notice boards and portals, an ordinary citizen rarely has a single, simple place to check, in their own language, exactly which schemes they qualify for, what benefit they would receive, and what documents they must carry. SachiSeva was built to close this gap: a free, bilingual Telugu/English web application, which can be added to a phone's home screen, where a family answers a few questions once, sees every scheme it is eligible for, and applies for all of them together.

## 2. Problem Statement

Citizens typically learn about a scheme only by asking Sachivalayam staff about one scheme at a time. Eligibility rules (income ceilings, age limits, caste/category criteria and asset limits) are spread across multiple notices and change whenever a scheme is revised, and the information is often in formal Telugu or English that is hard for citizens with limited literacy.

As a result, citizens frequently make repeat visits because a required document is missing, and submit the same documents again for each scheme. SachiSeva addresses these problems with a single bilingual tool that checks eligibility, combines document lists, and handles applications from submission to decision.

## 3. Objectives of the Project

The primary objective of the project is to give every family one bilingual place to discover, prepare for, apply for and track the welfare schemes it is entitled to. The specific objectives include:

- To let a citizen check, in one pass, which of the State's welfare schemes they are eligible for, in Telugu or English.
- To encode accurate, current eligibility rules for 21 schemes and 16 Sachivalayam services in 14 categories.
- To provide one combined document checklist with a Telugu/English voice read-out.
- To let a citizen apply for all eligible schemes in one step and follow each application.

- To give Sachivalayam staff a protected panel to review applications, request documents and record decisions.
- To protect personal data, so that each citizen sees only their own information.

## 4. Scope of the Project

SachiSeva covers 21 welfare schemes and 16 Sachivalayam services across 14 categories, including the schemes behind all six "Super Six" promises. It provides a one-click eligibility check for the whole family, personalised benefit amounts, a combined document checklist with voice read-out, online applications ("Apply for all"), application tracking with messages from staff, a nearest Sachivalayam/MeeSeva finder, the SachiBot assistant, and a staff panel protected by a staff key. Every page requires a login.

## 5. System Architecture

The application follows a three-tier architecture, as shown in Fig. 1. Citizens and staff use the same web application in their browser, which holds the bilingual catalogue and the eligibility engine, so a family's eligibility is worked out instantly on the phone. Supabase provides sign-in with separate citizen and staff roles, a PostgreSQL database whose row-level security rules ensure that each citizen can see only their own data, private storage for document photos, and live updates for status changes and messages. Vercel server functions connect to external services such as Google Gemini, so that secret keys never reach the browser.

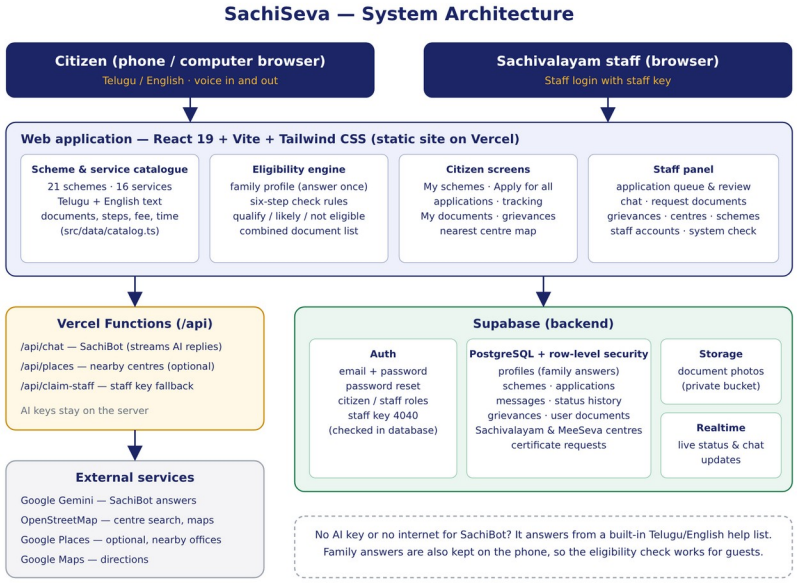



**Fig. 1** – System architecture of SachiSeva

## 6. Methodology

SachiSeva was developed using an iterative build-and-test approach. The scheme and service catalogue is kept in one source file in both languages and is used to generate the database entries and SachiBot's knowledge, so that all three always agree. All eligibility rules and scheme data were

verified manually against government sources, and every change was checked with automated database and browser tests.

**Table 5** – Stages of the Development Methodology

| Stage | Description |
|---|---|
| **Requirement Analysis** | Community survey and study of the Sachivalayam welfare-scheme system to identify essential features. |
| **System Design** | Architecture, scheme data model, eligibility engine, bilingual screens and security model. |
| **Development** | Citizen screens and staff panel in React; database, sign-in, storage and live updates on Supabase; server functions on Vercel. |
| **Data Verification** | Scheme names, benefits and documents checked against current government sources. |
| **Testing** | 65 automated database checks and browser tests of every screen in Telugu and English. |
| **Deployment** | Hosted on Vercel and demonstrated to the community. |

## 7. System Features

The application works in Telugu and English with a one-tap language switch. Its major features include:

- **Login first:** a log in / register screen for citizens and staff; every other page needs a login.
- **My schemes:** 23 short questions in three steps, including the government's six-step check, give eligible, likely and not eligible results with the personalised benefit.
- **One combined document list:** documents for all eligible schemes are merged, so each appears once.
- **Apply for all:** one form, each document uploaded once, and a separate application number for every scheme.
- **Applications and tracking:** a progress timeline, messages with staff, and an Upload button for every document staff request.
- **Nearest centre:** the closest Sachivalayam and MeeSeva centre with distance and directions.
- **SachiBot and read-aloud:** an assistant that answers in Telugu or English, and a speaker button on every page.
- **Staff panel:** review applications, change status, message citizens and ask for several documents at once. On approval or rejection the applicant is automatically asked to visit their nearby Sachivalayam.

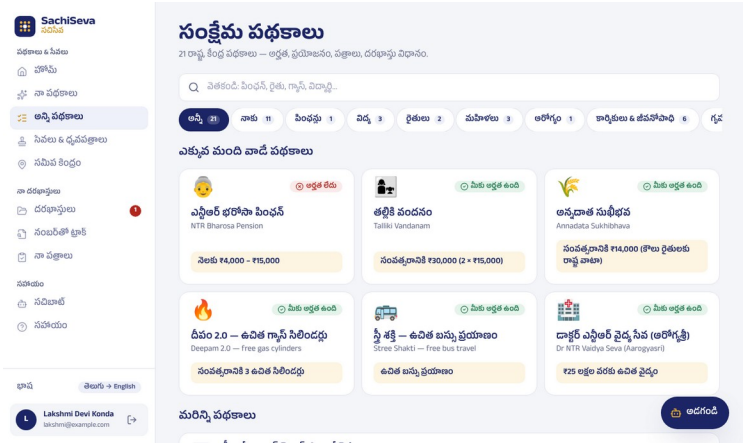



**Fig. 2** – Welfare scheme catalogue screen (Telugu)

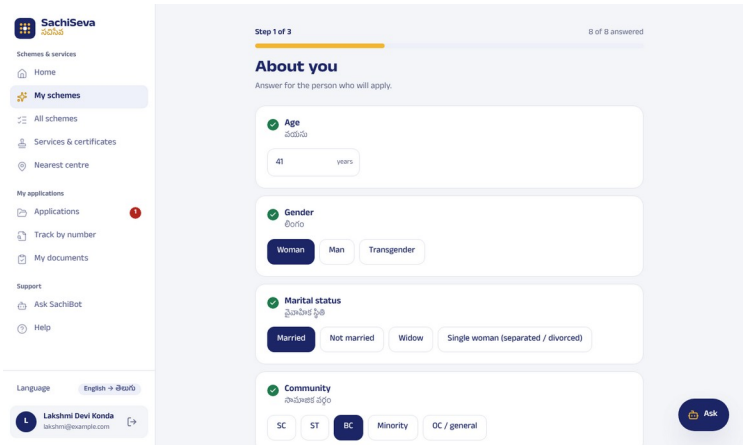



**Fig. 3** – Family questionnaire, answered once

## 8. Technologies Used

The successful implementation of the project required the integration of the following technologies.

**Table 6** – Technologies Used and Their Purpose

| Technology | Purpose |
|---|---|
| **React 19, TypeScript, Vite** | Single-page web application for citizens and staff |
| **Tailwind CSS** | Layouts that work on phones and desktops |
| **Supabase (PostgreSQL)** | Database with row-level security, sign-in, storage and live updates |
| **Vercel Functions** | SachiBot, nearby-office search and the staff-key check |
| **Google Gemini** | Answers for the SachiBot assistant |
| **OpenStreetMap and Leaflet** | Map and nearest Sachivalayam/MeeSeva centre |
| **Web Speech API** | Telugu/English voice read-out and voice input |
| **GitHub and Vercel** | Version control and hosting |

## 9. Results and Discussion

The completed application is deployed on Vercel. A citizen can open the link, answer the family questions once, and within seconds see exactly which of the 21 schemes they qualify for, the benefit amount and the documents required, collapsing several separate enquiries into one self-service check followed by a single "Apply for all" submission.

For example, a 67-year-old widow in a village with a family income of Rs. 6,000 a month is shown the NTR Bharosa pension at Rs. 4,000 a month, and a land-owning farmer family with two school-going children is shown Annadata Sukhibhava at Rs. 20,000 a year and Talliki Vandanam at Rs. 30,000 a year. 65 automated database checks confirmed that a citizen sees only their own data, cannot approve their own application, a wrong staff key is refused, and approving or rejecting an application sends the visit message. Browser tests opened all 37 screens on phone and desktop layouts in both languages without errors.

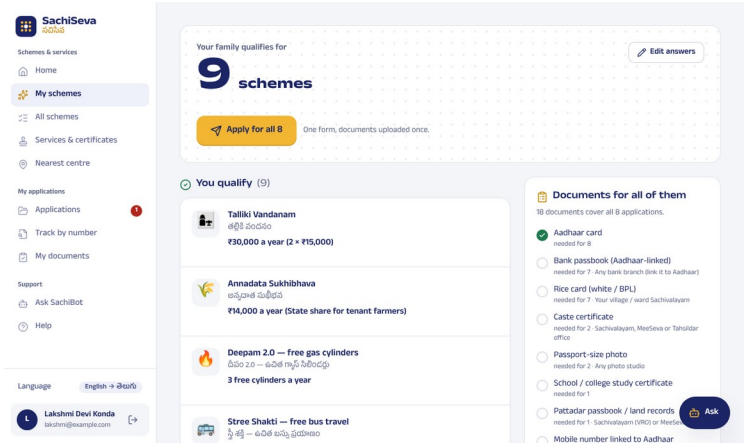



**Fig. 4** – My Schemes: eligibility results and combined document checklist

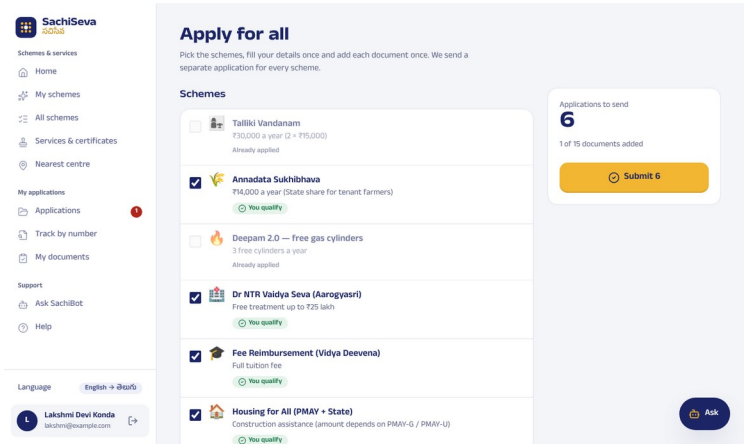



**Fig. 5** – Apply for all screen

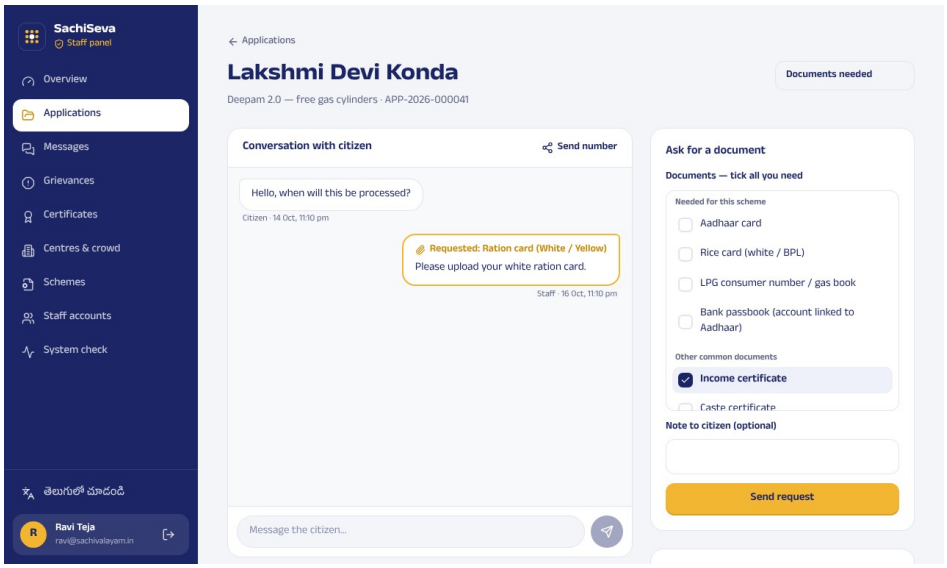



**Fig. 6** – Staff panel: application review

## 10. Challenges Faced

**Deployment:** the first version was built on a server-rendered framework that did not run correctly on Vercel, so it was rebuilt as a static React application with Vercel server functions.

**Scheme data accuracy:** scheme names, benefits and thresholds change whenever a programme is revised, so every entry was checked against real government sources, and announced-but-not-open schemes are clearly marked.

**Eligibility accuracy:** an early version marked every scheme as eligible because the stored rules were empty; the rules were rewritten so that each condition answers yes, no or "not known yet".

**Telugu text and voice:** Telugu did not render consistently until the font was bundled with the app, and not every browser ships a Telugu voice, which required testing on several devices.

**Security:** testing showed that a citizen could have created an application already marked as approved; database triggers and row-level security rules now prevent this.

## 11. Conclusion

SachiSeva shows that a citizen-facing e-governance tool can be both simple and complete. A bilingual catalogue, an eligibility engine based on the government's six-step check and a secure cloud database give an Andhra Pradesh family one place to discover every scheme it qualifies for, prepare its documents, apply for all of them together and follow each application to a decision. The project met its objectives: it covers 21 schemes and 16 services in 14 categories, offers a combined document checklist with Telugu voice read-out, and gives Sachivalayam staff a protected panel to review applications.

# CHAPTER 6: RECOMMENDATIONS AND CONCLUSIONS

**Recommendations:**

**Expand Scheme Coverage**

- Add new schemes and services as they are announced, and periodically re-verify benefit amounts and thresholds against official government sources.

**Government Partnership**

- Partner with the Sachivalayam system so that scheme updates and application status can be synced from official sources instead of manual entry.

**Push Notifications**

- Notify citizens about application updates and upcoming renewals without requiring them to open the app.

**Promote Digital Literacy**

- Train Sachivalayam staff and volunteers to demonstrate SachiSeva to visiting citizens as part of routine service delivery.

**Monitor and Evaluate**

- Track applications, approvals and repeat visits, and conduct a larger user study to measure the time saved per enquiry.

**Conclusions:**

SachiSeva – Welfare Eligibility Platform successfully addressed the welfare-scheme awareness gap among citizens served by the AP Sachivalayam system, bringing eligibility checking, document preparation, applications and tracking together in one bilingual web application.

Developed using React, TypeScript, Tailwind CSS, Supabase and Vercel, the system keeps personal data secure with row-level security, and was validated with 65 automated database checks and browser tests.

The survey at Dungalavanipalem, NAD Junction highlighted the need for a bilingual, self-service eligibility tool. Overall, SachiSeva proved to be a practical and accessible solution that also helped create awareness about government welfare schemes and the documents citizens need before visiting the Sachivalayam.

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# ANNEXURE A: FIELD PHOTOGRAPHS




Photo A.1 – The team discussing the project with a Sachivalayam official (15 June 2026)




Photo A.2 – The team at the Sachivalayam, Dungalavanipalem, NAD Junction (15 June 2026)




Photo A.3 – The team outside the Sachivalayam, Dungalavanipalem, NAD Junction (15 June 2026)




Photo A.4 – Demonstrating SachiSeva to a citizen on a phone (15 June 2026)

# ANNEXURE B: CSP COMPLETION CERTIFICATE

The following certificate was issued by the Secretary, Grama/Ward Sachivalayam, Dungalavanipalem (NAD), Visakhapatnam, on 15 June 2026, confirming the completion of the Community Service Project "SachiSeva" by the team during the academic year 2025–2026.



**GOVERNMENT OF ANDHRA PRADESH**

**GRAMA / WARD SACHIVALAYAM**

Dungalavanipalem, NAD Junction, Visakhapatnam, Andhra Pradesh - 530027

**COMMUNITY SERVICE PROJECT**
**COMPLETION CERTIFICATE**

This is to certify that the following B.Tech (CSE – Artificial Intelligence) students of Vignan's Institute of Information Technology (A), Visakhapatnam, have successfully completed their Community Service Project (CSP) titled **"SACHISEVA"** at this Sachivalayam during the academic year 2025–2026.

| | |
|---|---|
| **B. MAITHREYA** | – 24L31A4318 |
| **D. ABHILASH** | – 24L31A4340 |
| **J. MOUNIKA** | – 24L31A4357 |
| **K. YOGITHA** | – 24L31A4359 |

They have actively participated in the SachiSeva initiative, designing and developing a Telugu-first welfare scheme eligibility and assistance platform, and contributed sincerely towards promoting awareness of government welfare schemes among citizens. We appreciate their dedication, teamwork, and sense of social responsibility. We wish them success in all their future academic and professional endeavours.

WARD SECRETARIAT 1086473 ZONE - VI, GVMC

K.S. Kumary

Date : 15th JUNE 2026

Place : DUNGALAVANIPALEM (NAD)

**Signature of Secretary**

Grama/Ward Sachivalayam